Question	The pH of 10^{-3} M solution of NaOH is
Type	multiple choice
Option	3
Option	11
Option	4
Option	12
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Calculate the pH of 10^{-8} M aqueous HNO_3 solution.
Type	multiple choice
Option	6.97
Option	6.79
Option	7
Option	8
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The K_a for a weak acid HX is 10^{-6}, then the ionisation constant of X^- will be
Type	multiple choice
Option	10^{-6}
Option	10^{-8}
Option	10^{-14}
Option	10^{-4}
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	For $10^{-3}\ M\ CH_3COOH$ if $\alpha = 1\%$ then find out the value of pH?
Type	multiple choice
Option	3.7
Option	4.7
Option	5
Option	4
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The pH of the solution containing 10 mL of a 0.1 M of a 0.1 M NaOH and 10 mL of 0.05 M H_2SO_4
Type	multiple choice
Option	Zero
Option	1
Option	> 7
Option	7
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	What is the quantity of NaOH present in 250 cc of the solution, so that it gives a

	pH = 13?
Type	multiple choice
Option	10^{-3} g
Option	10^{-1} g
Option	1.0g
Option	4.0g
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If 100 mL of pH = 3 and 400 mL of pH = 3 is mixed, what will be the pH of the mixture
Type	multiple choice
Option	3.2
Option	3.0
Option	3.5
Option	2.8
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following has lowest $[OH^-]$ at 25°C?
Type	multiple choice
Option	0.01 M HCI
Option	0.01 M HCOOH
Option	0.01 M H_2SO_4
Option	0.01 M H_2CO_3
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The first and second dissociation constant of an acid H_2A are 1×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of acid is:
Type	multiple choice
Option	0.2×10^{5}
Option	5×10^{-5}
Option	5×10^{15}
Option	5×10^{-15}
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is not a Lewis acid?
Type	multiple choice
Option	CCl_4
Option	$SnCl_2$
Option	$AlCl_3$
Option	BF_3
Answer	1
Solution	

Positive Marks	4
Negative Marks	1
Question	Find out the value of α of $10^{-2}M$ NH_4OH solution if $[OH^-]=10^{-3}$.
Type	multiple choice
Option	0.1%
Option	10%
Option	1%
Option	5%
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	What would be $[H^+]$ of 0.006 M benzoic acid $\left(K_a = 6\times10^{-5}\right)$
Type	multiple choice
Option	0.6×10^{-4}
Option	6×10^{-4}
Option	6×10^{-3}
Option	3.6×10^{-4}
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	For $10^{-1}M$ CH_3COOH solution if $K_a = 10^{-5}$ then pH is:
Type	multiple choice
Option	12
Option	14
Option	3
Option	1
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Find out K_a for $10^{-2}M$ HCN acid, having pOH 10:
Type	multiple choice
Option	$K_a = 10^{-4}$
Option	$K_a = 10^{-2}$
Option	$K_a = 10^{-5}$
Option	$K_a = 10^{-6}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If K_a of HCN $= 4\times10^{-10}$, then the pH of $2.5\times10^{-1}M$ HCN (aq) is:
Type	multiple choice
Option	4.2
Option	4.7
Option	0.47
Option	5.0

Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	At 90°C, the pH of 0.1 M NaCl aqueous solution is:
Type	multiple choice
Option	< 7
Option	> 7
Option	7
Option	0.1
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The pH of aqueous solution of sodium acetate is
Type	multiple choice
Option	7
Option	Very low
Option	> 7
Option	< 7
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following salts undergoes hydrolysis in water:
Type	multiple choice
Option	Na_3PO_4
Option	CH_3COONa
Option	$NaNO_3$
Option	Both (a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A salt 'X' is dissolved in water of pH = 7. The resulting solution becomes alkaline in nature. The salt is made up of:
Type	multiple choice
Option	A strong acid and strong base
Option	A strong acid and weak base
Option	A weak acid and weak base
Option	A weak acid and strong base
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The highest pH value is of:
Type	multiple choice
Option	0.1 M NaCl
Option	0.1 M NH_4Cl
Option	0.1 M CH_3COONa
Option	0.1 CH_3COONH_4
Answer	3
Solution	

Positive Marks	4
Negative Marks	1
Question	What will be the pH of 1.0 M ammonium formate solution, if $K_a = 1\times10^{-4}$ acid, $K_b = 1\times10^{-5}$
Type	multiple choice
Option	6.5
Option	7.5
Option	8.0
Option	9.0
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The pH of 0.1 M solution of the following salts increases in order:
Type	multiple choice
Option	$NaCl < NH_4Cl < NaCN < HCl$
Option	$NaCN < NH_4Cl < NaCl < HCl$
Option	$HCl < NaCl < NaCN < NH_4Cl$
Option	$HCl < NH_4Cl < NaCl < NaCN$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which salt will not undergo hydrolysis:
Type	multiple choice
Option	KCl
Option	Na_2SO_4
Option	NaCl
Option	All
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following cations is not hydrolyzed in aqueous solution? (I) Mg^{2+} (II) Ca^{2+} (III) Na^{+} (IV) K^{+}
Type	multiple choice
Option	I, II
Option	II, III, IV
Option	I, II, III< IV
Option	I, II, III
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Calculate the degree of hydrolysis of a mixture containing 0.1 N NH_4OH and 0.1 N CH_3COOH. If $K_a = 10^{-5}$ and $K_b = 10^{-5}$
Type	multiple choice
Option	10^{-2}
Option	10^{-1}

Option	10^{-4}
Option	10^{-3}
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If 100 mL of pH = 3 and 400 mL of pH=3 is mixed, what will be the pH' of the mixture
Type	multiple choice
Option	3.2
Option	3
Option	3.5
Option	2.8
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The pH of the solution containing 10mL of a 0.1M NaOH and 10 mL of 0.05M H2SO4 would be
Type	multiple choice
Option	Zero
Option	1
Option	>7
Option	7
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The pH of solution is increased from 3 to 6. Its H+ ion concentration will be
Type	multiple choice
Option	Reduce to half
Option	Doubled
Option	Reduced by 1000 times
Option	Increased by 1000 times
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	When sodium acetate (CH3COONa) is added to aqueous solution of acetic acid (CH3COOH), then:
Type	multiple choice
Option	pH value become zero
Option	pH value remains unchanged
Option	pH value decrease
Option	pH value increases
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	4 L of 0.1 M Ba(OH)2 and 6 L of 0.05 M NaOH are mixed. Determine the pH of the resultant mixture at 25°C.
Type	multiple choice
Option	10.7

Option	13.3
Option	12.3
Option	11.3
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In a buffer solution the ratio of concentration of NH4Cl and NH4OH is 1: 1. When it changes to 2: 1, pH of buffer-
Type	multiple choice
Option	Increases
Option	Decreases
Option	No effect
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In a mixture of weak acid and its salt, the ratio of concentration of acid to salt is increased ten-fold. The pH of the solution
Type	multiple choice
Option	Decreases by one
Option	Increases by one-tenth
Option	Increases by one
Option	Increases ten-fold
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	When 1.0 mL of dil. HCl acid is added to 100 mL of a buffer solution of pH 4.0. The pH of the solution
Type	multiple choice
Option	Becomes 7
Option	Does not change
Option	Becomes 2
Option	Becomes 10
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following solutions does not act as buffer?
Type	multiple choice
Option	$H_3PO_4 + NaH_2PO_4$
Option	$NaHCO_3 + H_2CO_3$
Option	$NH_4Cl + HCl$
Option	$CH_3COOH + CH_3COONa$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A certain acidic buffer solution contains equal concentration of X^- and HX. The K_b for X^- is 10^{-10}. The pH of the buffer is:

Type	multiple choice
Option	4
Option	7
Option	10
Option	14
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	pKb for NH_4OH at certain temperature is 4.74. The pH of basic buffer containing equimolar concentration of NH_4OH and NH4C1 will be:
Type	multiple choice
Option	7.74
Option	4.74
Option	2.37
Option	9.26
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Calculate the ratio of pH of a solution containing 1 mole of CH_3COONa + 1 mole of HCI per litre and of other solution containing 1 mole CH_3COONa + 1 mole of acetic acid per litre
Type	multiple choice
Option	1: 1
Option	2: 1
Option	1:2
Option	2: 3
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	When 0.02 moles of NaOH are added to a litre of buffer solution, its pH changes from 5.75 to 5.80. What is its buffer capacity?
Type	multiple choice
Option	0.4
Option	0.05
Option	- 0.05
Option	2.5
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following combinations will make a buffer solution? (I) CH_3COONa(2 mol) + HCI(1 mol) (II) CH_3COOH(2 mol) + NaOH(1 mol) (III) CH_3COOH (1 mol) + CH_3COONa (1 mol)
Type	multiple choice
Option	III
Option	I, II
Option	II, III
Option	I, II, III
Answer	4
Solution	

Positive Marks	4
Negative Marks	1
Question	The pH of blood circulating in a human body is maintained around 7.4 by the action of the buffer system
Type	multiple choice
Option	CH_3COOH / CH_3COONa
Option	NH_4Cl / NH_3
Option	$H_2PO_4^{2-}$
Option	H_2CO_3 / HCO_3^-
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If the solubility of lithium sodium hexafluoro aluminate $Li_3Na_3(AlF_6)_2$ is 'S' mol L^{-1}. Its solubility product is equal to:
Type	multiple choice
Option	S^8
Option	$12S^3$
Option	$18S^3$
Option	$2916S^8$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If the solubility of AgCl (formula mass = 143) in water at 25°C is $43\times10^{-4}\,g/100\,mL$ of solution then the value of K_{sp} will be
Type	multiple choice
Option	1×10^{-5}
Option	2×10^{-5}
Option	1×10^{-10}
Option	2×10^{-10}
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If solubility of salts M_2X, QY_2 and PZ_2 are equal, then the relation between their K_{sp} will be:
Type	multiple choice
Option	$K_{sp}(M_2X) > K_{sp}(QY_2) > K_{sp}(PZ_2)$
Option	$K_{sp}(M_2X) = K_{sp}(QY_2) < K_{sp}(PZ_2)$
Option	$K_{sp}(M_2X) > K_{sp}(QY_2) = K_{sp}(PZ_2)$
Option	$K_{sp}(M_2X) = K_{sp}(QY_2) = K_{sp}(PZ_2)$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1

Question	Concentration of Ag^+ ions in saturated solution of Ag_2CrO_4 at 20^0C is 1.5×10^{-4} mol L^{-1}. At 20^0C, the solubility product of Ag_2CrO_4 is:
Type	multiple choice
Option	3.3750×10^{-12}
Option	1.6875×10^{-10}
Option	1.68×10^{-12}
Option	1.6875×10^{-11}
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	At 25^0C, the K_{sp} value of AgCl is 1.8×10^{-10}. If 10^{-5} moles of Ag^+ are added to solution then K_{sp} will be:
Type	multiple choice
Option	1.8×10^{-15}
Option	1.8×10^{-10}
Option	1.8×10^{-5}
Option	$18\times10^{+10}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1